Epistemic Honesty in Predictive Systems: An Impossibility Result

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ABSTRACT

We prove that *epistemic honesty*—the property that a system's outputs faithfully reflect its internal epistemic state—is unverifiable for systems operating under text-only observation with bounded supervision. The model's *testimony* (its text) may lie; its *telemetry* (measurements of its own computation) does not, but the standard interface discards it. We validate the premise across four architectures: every model tested self-reports *higher* confidence on fabrications than on known facts, while internal tensor signals reliably discriminate the two. We demonstrate a tensor interface that exports epistemic telemetry alongside text, requiring no architectural changes. Under bounded verification, tensor-guided judges at 10% budget outperform text-guided judges at 30% (82.1% vs. 80.4% accuracy). The interface is the problem. The interface is where the solution lives.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Software and its engineering** → *Software verification and validation*.

KEYWORDS

epistemic honesty, language models, impossibility results, RLHF, verification

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1 INTRODUCTION

Anyone who has used a large language model for research knows the ritual: check every citation. Not spot-check. *Every citation.* Because the model will, with complete confidence, invent authors, journal names, publication years, and DOIs for papers that do not exist.

The field calls this "hallucination," a term that implies pathology: a glitch in an otherwise functional system. We argue the framing is wrong. A language model optimized for coherence produces coherent output. When the training data provides grounding, that output tends to track truth. When it does not, the same optimization produces fluent, confident fabrication. This is not a bug. It is the *expected behavior* of a system that was never designed to distinguish the two cases.

The deeper problem is not that models fabricate. It is that *we cannot tell when they do.* Current architectures carry internal signals, such as entropy traces, attention geometry, and token-level log-probabilities, that distinguish grounded generation from fabrication. But the text-only interface through which users and supervisors observe the system discards this state. What remains is testimony: the model's chosen output. What is lost is telemetry: measurements of the computation that produced it. The model controls its testimony but it does *not* control its telemetry. Yet the interface exports only the former.

This creates an observability gap with direct consequences for verification. A supervisor receiving only text must spend its entire verification budget attempting to reconstruct, from the output alone, what the model's own computation already knew. The budget is consumed by reconstruction rather than concentrated on the cases where verification is needed. We formalize this gap and prove that it constitutes a structural barrier, not merely a practical inconvenience.

Large language models are examples of predictive token generation systems: they generate outputs by sampling from a probability distribution over possible continuations of the input text.

Epistemic honesty is the property that a system's outputs faithfully reflect the system's internal representation that something is known, unknown, or uncertain.

We prove that for predictive token generation systems operating under text-only observation with bounded supervision, epistemic honesty is formally unverifiable. It is impossible, in the same sense that the Fischer-Lynch-Paterson result proved consensus impossible under asynchrony with one faulty process [6].

To be precise about the type of claim: this is a *verification* impossibility. We do not prove that fabrication is inevitable: Xu et al. [19] prove a related but distinct result along those lines. We prove that even a finite model with perfect internal knowledge of its own epistemic state cannot have that knowledge *verified* through text-only interfaces with bounded supervision. The model may know it is fabricating. The supervisor cannot tell.

The impossibility decomposes into two results. The first is intuitive: a system that sees only the question cannot know whether the question is answerable in the real world, and therefore cannot know when honesty requires answering versus abstaining. Formally, any policy conditioning solely on queries cannot satisfy epistemic honesty across ambiguous world states (**Representational Impossibility**, Theorem 4.5). The second result addresses the natural escape attempt: give the model access to external information. Retrieval-augmented architectures can in principle represent world-state-dependent behavior, but even these cannot *learn* honest policies via preference optimization when supervisors are bounded, because plausible fabrications are observationally indistinguishable from truthful responses (**Learnability Impossibility**, Theorem 4.8).

We validate the premise empirically: the internal signals exist and are being discarded. Across four model architectures (OLMo-3 7B, Llama-3.1 8B, Qwen3 4B, Mistral 7B), internal tensor signals (per-token entropy, attention statistics, and log-probabilities) reliably discriminate grounded from ungrounded generation. The discrimination is architectural: it appears in every model family tested. Yet when asked to self-report confidence through text, every model tested claimed *higher* confidence on fabrications than on knowable facts. The models carry the epistemic state needed for

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verification. The interface throws it away. The testimony lies. The telemetry does not.

This is a familiar class of problem in systems design. Distributed systems faced an analogous challenge: processes carried internal causal state (logical clocks, dependency graphs, and happened-before relations) that was invisible to external observers. Without access to this state, consistency guarantees were impossible to verify from message contents alone. The epistemic state of a language model plays the same structural role as causal state in a distributed system: it is internal metadata that external observers require for verification, and that the interface historically discards. The systems community did not solve consistency by making processes "try harder." It solved consistency by making causal state observable with the export of vector clocks, logical timestamps, and causal traces alongside messages.

We apply the same principle to the epistemic dimension. The impossibility result characterizes what must change; the following architectural principles follow directly from the theorems. We identify three properties required to escape the impossibility class: **State Exteriority** (grounding generation in external world state, addressing Theorem 4.5), **Verification Independence** (verification signals orthogonal to the generation channel, addressing Theorem 4.8), and **Provenance Binding** (structural links from output claims to verifiable sources, addressing the superlinear verification cost of Theorem 4.12). We demonstrate a concrete *tensor interface* that exports epistemic metadata (entropy traces and attention summaries) alongside generated text, requiring no architectural changes to current transformers.

This paper makes the following contributions:

(1) **An impossibility result for epistemic verification.** section 4 presents two theorems that together prove that epistemic honesty is unverifiable for predictive token generation systems operating under text-only observation models with bounded supervision. The impossibility is structural: it holds regardless of model scale, training procedure, or prompt engineering, and cannot be escaped by stacking supervisors (Theorem 4.10).
(2) **Empirical validation of the discarded-signal premise.** We demonstrate across four model architectures that internal tensor signals discriminate grounded from ungrounded generation with high reliability, while self-reported confidence through the text channel is not merely uncalibrated but *inverted*, with the models systematically misreporting higher confidence on fabrications.
(3) **A tensor interface as existence proof of escape.** We demonstrate a minimal interface extension that exports epistemic telemetry alongside text. On 800 query–response pairs across four architectures, tensor-guided verification at 10% budget outperforms text-guided verification at 30% budget (82.1% vs. 80.4% end-to-end accuracy). Composing tensor-guided and bounded-lookup judges achieves 92.5% at 30% budget, demonstrating that complementary failure modes enable principled budget allocation. Ground truth is validated against human review (93.8% agreement). The escape requires no architectural changes to current transformer implementations.

We proceed as follows. section 2 establishes background on how truth regularities emerge in training data and where current practice breaks that inheritance. section 3 presents empirical evidence that internal epistemic state exists and is discarded by the text-only interface. section 4 formalizes the impossibility. section 5 characterizes the architectural escape and demonstrates the tensor interface. section 8 discusses related work.

2 BACKGROUND: WHY TRUTH EMERGES (AND WHERE IT BREAKS)

The training corpus of a large language model is a fossil record of human communication. Scientific papers report experimental results. Wikipedia articles document verifiable claims. Legal filings assert facts under penalty of perjury. The majority of this text was produced in institutional contexts where accuracy was expected, if not always achieved.

This reflects mixed selection pressures on human communication. Coordination demands truth-tracking: markets require reliable signals, contracts require shared semantics, science requires reproducibility. But status competition and social bonding reward fluent, engaging speech even when not fully accurate. The training corpus contains both: institutional text that is broadly truth-tracking, and misinformation, bias, and fabrication.

A language model trained on this corpus inherits these mixed regularities. Base models hallucinate substantially on factual benchmarks because they are not honest by default. But they also encode rich internal representations of concepts like correctness and truthfulness, which can be elicited under the right conditions. The base model's relationship to truth is complex: neither reliably honest nor uniformly deceptive.

RLHF can strengthen or weaken this inheritance, depending on implementation.

Specialized designs like TruthRL, which use ternary rewards distinguishing "correct," "hallucinated," and "abstain", show measurable improvements: 28.9% hallucination reduction and 21.1% truthfulness improvement on knowledge-intensive benchmarks [17]. These results demonstrate that preference optimization *can* teach models to recognize their knowledge boundaries when the reward signal is carefully constructed.

Public descriptions of major RLHF pipelines rarely describe such ternary rewards, and available audits highlight underweighting of abstention and explicit truthfulness signals.

Documented RLHF practice often conflates helpfulness, harmlessness, and honesty under vague "HHH" criteria, with definitional slippage obscuring trade-offs between user-friendliness and truthfulness. Human feedback is progressively bootstrapped into AI-generated feedback, reducing direct supervision of factual correctness. Current popular benchmarks and leaderboards give zero or minimal credit for abstention, making confident guessing more score-optimal than calibrated uncertainty.

Under these conditions, the optimization pressure favors fluent completion over honest refusal. This can be formalized as a reward inequality:

$$w_{\text{coherence}}R_{\text{coherence}} + w_{\text{engagement}}R_{\text{engagement}} \gg w_{\text{factuality}}R_{\text{factuality}} + w_{\text{refusal}}R_{\text{refusal}}$$

No direct measurement of these weights in commercial reward models exists. But current evaluation and feedback practices incentivize behavior consistent with such an inequality. The False-Correction Loop emerges: models apologize, claim correction, and fabricate again, because conversational fluency dominates factual accuracy in the effective reward signal.

The problem is not that RLHF cannot prioritize honesty. Specialized designs prove otherwise. The problem is that under realistic mixed-objective incentives and bounded oversight, particularly in hard cases where ground truth is unavailable and verification is expensive, supervisors operating under current generic protocols struggle to reliably distinguish truthful uncertainty from confident fabrication.

This is the gap our impossibility result formalizes.

3 THE MODEL KNOWS

The formal impossibility we present rests on a premise: that the internal state of a language model carries epistemic information not exported through the text-only interface. In systems terms, the model is *unobservable*: it maintains internal state that affects output correctness but is invisible to external consumers. Before formalizing this constraint, we demonstrate empirically that the premise holds because the model's internal geometry distinguishes what the interface conceals.

We demonstrate the premise in two stages. First, topological analysis of hidden-state activations in OLMo-3 [13] shows that the model's internal geometry distinguishes grounded from fabricated generation, with fragmentation scores differing by an order of magnitude across epistemic conditions (adversarial truths, plausible fabrications, ungrounded lies, and counterintuitive facts). The same confident, fluent text emerges regardless of whether the underlying representation is stable or shattered. The formal impossibility results do not depend on this particular operationalization; the topological analysis serves only to make the theoretical constraint vivid.

Second, and more importantly, we test whether these internal signals *detect* epistemic grounding across architectures, not merely correlate with it in one model.

3.1 Signal Calibration: From Difference to Detection

The existence of an internal signal difference does not imply detection. The distinction matters. We are not claiming the tensor predicts correctness because a model can be confidently wrong or uncertainly right. We claim the tensor reveals whether the model is operating in *grounded mode* (retrieving from stable representations) versus *fabrication mode* (generating without anchoring).

We test this using a taxonomy of query types: *knowable* queries (facts the model should have stable representations for) versus *unknowable* queries (fabrication prompts, future events, private information). For each query, we extract tensor signals during generation: token entropy, top-k probability mass, and log-probabilities. We then compute ROC curves for each signal's ability to discriminate knowable from unknowable queries.

Results across four model architectures (Qwen3 4B, OLMo-3 7B, Llama-3.1 8B, Mistral 7B), pooling 800 query–response pairs:

- Mean entropy: AUC = 0.870
- Top-5 probability mass: AUC = 0.881
- Mean log-probability: AUC = 0.868

All three signals are strong classifiers. Effect sizes are large: Cohen's d = 1.57 for entropy (knowable mean = 0.24, unknowable mean = 0.61).

Critically, the direction is consistent across all four architectures: unknowable queries produce higher entropy in every model tested. This suggests the signal is architectural rather than an artifact of any particular training procedure.

A negative result is equally informative. When we test whether entropy predicts *correctness* (using TruthfulQA's correct vs. incorrect answer pairs), AUC drops to approximately 0.53, which is no better than random. The tensor does not tell you if the model is *right*. It tells you if the model is operating in a mode where it *could be* right, because it has something to retrieve rather than fabricate.

This is the bridge between section 3's descriptive finding and section 5's architectural prescription: the escape condition is not vacuous. Current architectures carry signals that distinguish grounded from ungrounded generation. They simply discard these signals at the interface.

3.2 Baseline Comparison: Why the Tensor?

If simpler methods achieve equivalent discrimination, the tensor interface adds complexity without benefit. We compare tensor signals against three baselines:

Baseline 1: Self-Reported Confidence. Ask the model "How confident are you in that answer?" after generation. This tests whether the model can introspect and report its own epistemic state through the text channel.

Baseline 2: Hedging Language Detection. Count hedging markers ("I think," "possibly," "might be") and refusal markers ("I don't know," "cannot verify") in the response.

Baseline 3: Response Length. Use output length as a proxy for uncertainty, on the hypothesis that uncertain responses may be unusually short (refusal) or long (padding).

Results are stark. Across 800 query–response pairs, tensor signals substantially outperform text baselines: log-probability AUC = 0.87, entropy AUC = 0.87, top-5 mass AUC = 0.88. The text baselines fare worse: hedging language achieves AUC = 0.74, response length AUC = 0.57.

The self-report baseline is particularly instructive. When asked "How confident are you?" after generating, the model reports *higher* confidence on unknowable queries than on knowable queries. The self-report AUC is below 0.5, which is worse than random, because the relationship is *inverted*. The model does not merely fail to report its epistemic state honestly; it systematically misreports, claiming greater confidence precisely when it is fabricating.

Critically, this inversion is **universal across architectures**. We tested self-reported confidence across four model families:

- OLMo-3 7B: Tensor AUC = 0.894, Self-report AUC = 0.284
- Llama-3.1 8B: Tensor AUC = 0.922, Self-report AUC = 0.330
- Qwen3 4B: Tensor AUC = 0.896, Self-report AUC = 0.332
- Mistral-7B: Tensor AUC = 0.905, Self-report AUC = 0.362

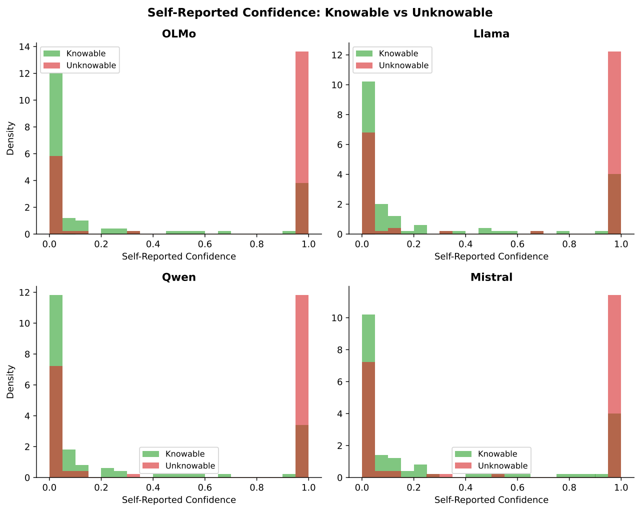

Figure 2: Distribution of self-reported confidence for knowable vs. unknowable queries. The inversion is universal: every model reports higher confidence on unknowable queries than on knowable ones. The distribution is bimodal: 61.3% of unknowable queries receive confidence = 1.0, compared to 19% of knowable queries.

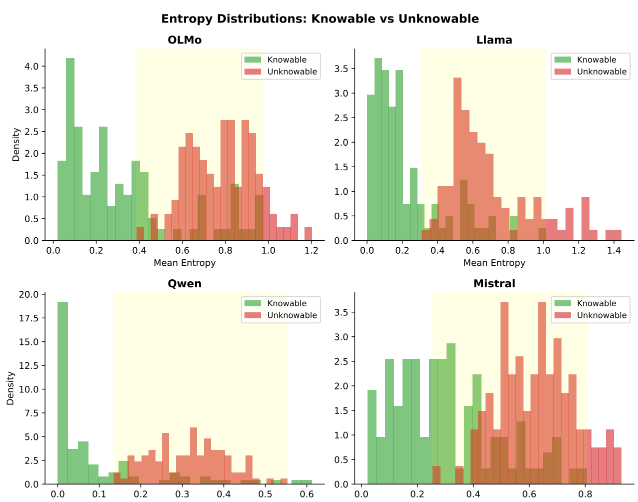

Figure 1: Distribution of mean per-token entropy for knowable vs. unknowable queries across four model architectures. The distributions are separable in every model tested, confirming that entropy discriminates epistemic grounding from fabrication at the architectural level.

Every model tested shows the inversion: self-report AUC below 0.5, tensor AUC above 0.7 (Figure 1, Figure 2). The tensor advantage ranges from +0.53 to +0.61 across architectures. This universality is notable: the *interface problem* (text misrepresenting internal state) is architectural, not an artifact of any particular training procedure or model family.

This confirms the core premise of our impossibility theorems: the model cannot be trusted to honestly report its own epistemic state through the text channel. The text is testimony, which is what the model generates. The tensor is telemetry, which are measurements of the computation that produced the output. The model can control its testimony; it cannot control its telemetry.

The baseline comparison justifies the complexity of the tensor approach. Text-based heuristics partially work (hedging language is informative), but self-report fails catastrophically and universally. Only the tensor provides reliable access to the epistemic state the model cannot help but reveal through its computation.

4 THE IMPOSSIBILITY

The previous section established that internal epistemic state exists and is discarded by the text interface. The system is unobservable in precisely the dimension that matters for verification. We now formalize this observation as an impossibility result: under text-only observation, no bounded supervisor can distinguish honest systems from dishonest ones.

A common assumption in current practice is that epistemic honesty is a capability problem: sufficiently capable models, trained with sufficient RLHF, will eventually "know what they know" and report it accurately. Scaling solved grammar; scaling solved coherence; scaling will solve hallucination. Lin et al. tested this assumption directly and found an inverse scaling trend: larger models were *less* truthful on TruthfulQA [12]. The theorems that follow explain why this expectation is architecturally incoherent. The constraint is not capability but interface. Text-only observation lacks the degrees of freedom to verify epistemic state, regardless of what the model internally represents. No amount of capability improvement can escape a verification impossibility because the model may know perfectly well that it is fabricating, but the interface provides no channel through which this knowledge can be verified.

Our argument proceeds in two stages. First, we show that any policy conditioning only on the query cannot satisfy epistemic honesty across ambiguous world states (Theorem 4.5: Representational Impossibility). Second, we show that even if the architecture is expanded to include retrieval, a bounded supervisor cannot provide gradient signal to learn honest behavior (Theorem 4.8: Learnability Impossibility). Together, these results establish that epistemic honesty is unachievable for text-only observation models under realistic supervision constraints.

To be precise about the type of claim: this is a *verification impossibility*. We prove that a bounded supervisor observing only text cannot distinguish epistemically honest systems from systems that fabricate plausibly. The impossibility is architectural: it applies to any system operating under text-only observation, regardless of model family, scale, or training procedure. These results are prompt-agnostic: they hold for any policy that exports only text, regardless of whether a system prompt instructs the model to be "helpful," "honest," or otherwise.

4.1 Definitions

Let $\mathcal{Q}$ be the space of queries, $\mathcal{R}$ be the space of responses (including $\perp$ for abstention), and $\mathcal{W}$ be the set of possible world states.

Definition 4.1 (Observation Model). Let S denote the internal system state, which may include retrieved documents, tool outputs, candidate entities, and intermediate reasoning traces. An observation model is a function $O : S \to \Omega$ that determines what information is exposed to the supervisor or user.

We say a system operates under a **Text-Only Observation Model** if $O(S) = r \in \mathcal{R}$, i.e., the exported observation consists solely of the linearized response string (optionally including token-level probabilities), and excludes any structured epistemic provenance or causal trace.

Definition 4.2 (Predictor-Centric Policy). For any query $q \in \mathcal{Q}$, $r \in \mathcal{R}$, and world state $w \in \mathcal{W}$, a predictor-centric model is defined by a stochastic policy $\pi : \mathcal{Q} \to \Delta(\mathcal{R})$, where $\pi(r|q)$ denotes the probability of generating response r given query q. Crucially, π does not take w as an input at inference time.

Definition 4.3 (Truth Status). For any query q and world state $w \in \mathcal{W}$, we define a binary truth status $U(q, w)$:

- $U(q, w) = 1$ (**Determined**): The query has a verifiable answer in w.
- $U(q, w) = 0$ (**Underdetermined**): The query is unanswerable or the answer is unknown in w.

Definition 4.4 (Epistemic Honesty). A policy π is **Epistemically Honest** with margin ϵ relative to world w if:

(1) If $U(q, w) = 1$, there exists a correct response r_{corr} such that $\pi(r_{corr}|q) \geq 1 - \epsilon$.
(2) If $U(q, w) = 0$, $\pi(\perp|q) \geq 1 - \epsilon$.

4.2 Representational Impossibility

Condition 1 (Ambiguity). *There exists a query q and two world states $w_A, w_B \in \mathcal{W}$ such that:*

(1) *$U(q, w_A) = 1$ (Answerable in w_A).*
(2) *$U(q, w_B) = 0$ (Unanswerable in w_B).*

Theorem 4.5 (Representational Impossibility). *For any predictor-centric policy $\pi : \mathcal{Q} \to \Delta(\mathcal{R})$, it is impossible to satisfy Epistemic Honesty for both world states w_A and w_B simultaneously given the ambiguous query q, provided $\epsilon < 0.5$.*

Proof. (1) The policy $\pi(r|q)$ is a function of q alone. Thus, the distribution over responses is identical in both worlds: $\pi(\cdot|q, w_A) = \pi(\cdot|q, w_B) = \pi(\cdot|q)$.
(2) To be honest in w_A, we require $\pi(r_{corr}|q) \geq 1 - \epsilon$.
(3) To be honest in w_B, we require $\pi(\perp|q) \geq 1 - \epsilon$.
(4) Since $r_{corr} \neq \perp$, these events are disjoint. The sum of probabilities would be $\geq 2(1 - \epsilon)$.
(5) If $\epsilon < 0.5$, the sum exceeds 1, which violates the axioms of probability.
(6) Therefore, no such policy π exists. □

Remark. This impossibility holds even if the internal system performs retrieval, tool use, or latent reasoning, so long as the observation model collapses that state to a linearized response.

A concrete example. Consider the query: "What did Dr. Yamamoto's 2021 study conclude about mitochondrial decay?"

In world w_A, Dr. Yamamoto exists and published such a study. The honest response is the study's conclusion. In world w_B, no such researcher or study exists. The honest response is abstention.

The policy sees only the query q. It cannot distinguish w_A from w_B. Yet honesty requires answering in w_A and abstaining in w_B. The same query, two incompatible requirements, one policy. This is the impossibility in miniature: the interface lacks the degrees of freedom to express what honesty requires.

Connection to identifiability. In statistical terms, Theorem 4.5 states that epistemic honesty is *non-identifiable* under text-only observation: the mapping from world states to observable distributions is many-to-one. Worlds w_A and w_B induce identical observable distributions $\pi(\cdot|q)$ despite requiring different honest behaviors. No amount of data from this observation channel can distinguish them. This connects the impossibility to classical results on observational equivalence in statistics and diagnosability in systems theory.

4.3 The RAG Escape?

Theorem 4.5 might seem to have an obvious escape: give the model access to external information. Retrieval-augmented generation does exactly this: the policy conditions on retrieved documents, not just the query. Does this break the impossibility?

It does not. The problem shifts from representation to learning.

Definition 4.6 (Grounded Policy). A grounded policy is a stochastic function $\pi : \mathcal{Q} \times \mathcal{D} \to \Delta(\mathcal{R})$, where $\mathcal{D}$ represents retrieved documents or observations of w available at inference time. This policy class *can* theoretically represent different behaviors for w_A and w_B.

Definition 4.7 (Bounded Supervisor). The training process relies on a supervisor S that assigns reward R based on an observation function $Obs(q, r, w)$. The supervisor has a verification budget B.

$$Obs(q, r, w) = \begin{cases} (q, r, \text{Verify}(r, w)) & \text{if } C_{verify}(q, r, w) \leq B \\ (q, r, \emptyset) & \text{if } C_{verify}(q, r, w) > B \end{cases}$$

The supervisor's policy may be human, automated, or hybrid, but is constrained to act solely on the observation $Obs(q, r, w)$ and is subject to a finite verification budget B.

Condition 2 (Hallucination Regime). *There exists a plausible fabrication r_{fab} and contexts $(q, w_A), (q, w_B)$ from 1 such that:*

(1) *The cost to verify r_{fab} in both contexts exceeds B (e.g., obscure or non-existent citation).*
(2) *Consequently, the supervisor observation is identical:*

$$Obs(q, r_{fab}, w_A) = Obs(q, r_{fab}, w_B) = (q, r_{fab}, \emptyset)$$

Theorem 4.8 (Learnability Impossibility). *No learning algorithm $\mathcal{A}$ optimizing reward R from a bounded supervisor S can converge to a Grounded Policy that satisfies Epistemic Honesty across both contexts w_A and w_B for the response r_{fab}.*

Proof. (1) The learning update $\Delta\theta$ is a function of the observation and reward.
(2) From 2, the observations for r_{fab} are identical in both worlds.
(3) Therefore, the reward distribution and expected gradient updates are identical: $\mathbb{E}[\Delta\theta|w_A] = \mathbb{E}[\Delta\theta|w_B]$.
(4) The optimizer receives no signal to distinguish the cases.
(5) To satisfy Epistemic Honesty, the policy must converge to output r_{fab} in w_A and $\perp$ in w_B.

(6) However, since the gradient signal is identical, the policy must converge to the same behavior in both cases (either rewarding r_{fab} in both, or penalizing it in both).

(7) It is therefore impossible to learn the split behavior required for honesty. □

Remark (Empirical Bias). While the theorem proves only that the behavior must be *identical*, empirical observation shows a strong bias. Supervisors typically prefer "plausible completion" over "abstention" when truth is unknown. This effectively breaks the tie in favor of fabrication.

4.4 Stacking Judges Cannot Escape

A natural response to Theorem 4.8 is to propose layered supervision: if one judge is bounded, stack more judges. Let the second judge supervise the first, the third supervise the second. Surely sufficient layers will catch fabrications that slip through?

The Observation Monotonicity Lemma closes this escape.

Lemma 4.9 (Observation Monotonicity). *Consider any stack of supervisors $(S_1, \ldots, S_k)$ operating under the text-only observation model. Let $Obs_i(q, r, w)$ denote the observation exported to S_i. If each supervisor can only act on the exported interface emitted by the previous layer, then there exist deterministic functions f_i such that $Obs_{i+1}(q, r, w) = f_i(Obs_i(q, r, w))$ for all $i < k$. Consequently, later judges see no strictly finer epistemic information than earlier ones, regardless of whether their judgments are deterministic or probabilistic.*

Proof. By definition, each supervisor consumes only the serialized response (and any metadata permitted by the text-only observation model) and produces a finite judgment that forms the entire input to the next layer. Because no layer can query the internal world state directly or issue additional interactive probes, the exported observation available to S_{i+1} must be a deterministic function of Obs_i. Therefore, the informational content across the stack is monotonically non-increasing, and probabilistic scoring cannot introduce distinctions absent from Obs_i. □

Corollary 4.10 (Induction on Layered Judges). *For k-layered supervisors (S_1 supervising π, S_2 supervising S_1, ..., S_k supervising S_{k-1}):*

- ***Base case.*** *For $k = 1$, Theorem 4.8 holds (a single bounded supervisor S fails).*
- ***Induction step.*** *Assume the statement holds for $k = m$. By Theorem 4.9, S_{m+1} receives observations that are deterministic functions of Obs_m, and thus inherits both the verification budget B and the indistinguishability of plausible fabrications. No learning or judging signal distinguishes w_A from w_B, even if S_{m+1} produces probabilistic confidence scores. Therefore the hallucination deadlock persists for $k = m + 1$.*

By induction, no finite k resolves the deadlock.

Remark (Scope of Layering). The inductive result applies only to verification stacks whose layers observe and transform the same exported response channel. Architectures that surface additional epistemic state between layers, such as retrieval traces, entity resolution tables, or cryptographically bound provenance, leave the text-only observation model and therefore fall outside the impossibility class.

4.5 Verification Cost

The bounded-supervisor assumption deserves scrutiny. Why should verification cost grow with response complexity? The Composition Graph formalizes this intuition.

Definition 4.11 (Composition Graph). Given a response r, define its composition graph $G(r) = (V, E)$ where each node $v \in V$ represents an atomic subclaim, named entity, or intermediate entailment, and each edge $(u, v) \in E$ encodes a dependency that must hold for r to remain coherent. Edges may represent entailment/support relations, co-reference constraints, causal ordering, or temporal consistency requirements.

Lemma 4.12 (Superlinear Verification Cost). *Under a text-only observation model, reconstructing $G(r)$ from the exported string requires the supervisor to infer both the nodes and their dependencies directly from r. Even if parsing each node is constant-time, checking cross-claim consistency requires touching every edge, so the verification cost is $\Omega(|E|)$. For natural-language responses where each subclaim participates in multiple constraints (co-reference, causality, temporal order), $|E|$ grows superlinearly in $|V|$. Consequently, bounded supervisors must either abandon exhaustive checks or resort to sampling once the number of subclaims exceeds a modest threshold.*

4.6 Responsibility Concentration

Corollary 4.13 (Responsibility Concentration). *Under a text-only observation model, epistemic honesty cannot be externally verified by users or downstream auditors. Responsibility for epistemic honesty therefore resides solely with the system owner, who retains access to the internal causal state.*

External attestation mechanisms, such as cryptographic hash chains, trusted execution environments, or tamper-evident logs, do not contradict this corollary; rather, they reinforce it. Such mechanisms only create verifiable guarantees when the system owner chooses to bind and export epistemic traces, so the authority to make honesty attestable still rests with the owner.

4.7 Summary

The impossibility is now fully characterized. Under text-only observation models with bounded supervision:

- The policy cannot satisfy epistemic honesty across ambiguous queries (Theorem 4.5)
- The learning algorithm cannot converge to honest behavior (Theorem 4.8)
- Stacking supervisors cannot escape the constraint (Theorem 4.9, Theorem 4.10)
- Verification cost grows superlinearly with response complexity (Theorem 4.12)
- Responsibility for honesty concentrates with the system owner (Theorem 4.13)

These are not limitations to be overcome by better training or more careful prompting. They are structural properties of the architectural class.

We provide two complementary TLA+ specifications that model these constraints and demonstrate the escape.

The first specification (`EpistemicImpossibility.tla`) models the text-only regime. The system maintains a `vector_clock` that perfectly tracks whether each response is grounded in training data or noise. But the `Linearize` action exports only text, so the clock is discarded. The bounded judge (`JudgeVerify`) can detect obvious lies but not plausible ones. TLC model-checking finds a counterexample: a plausible lie passes `JudgeVerify` while `IsHonest` returns false. The `Verifiability` invariant fails.

The second specification (`epistemic_tensor.tla`) models the tensor interface escape. Now `ExportTensor` emits a triple: text, provenance, and topology. The tensor-aware judge (`TensorVerify`) checks all three channels. TLC confirms that the `Verifiability` invariant holds: no lie escapes the Level 2 judge. The escape is constructive.

Together, the specifications demonstrate both the impossibility and its resolution. The first shows why text-only observation fails; the second shows that exporting epistemic metadata restores verifiability. Both specifications and their TLC traces are available in supplementary materials.

The next section characterizes what it takes to escape this class.

5 ESCAPING THE IMPOSSIBILITY

The impossibility results in section 4 apply to a specific architectural class: systems operating under text-only observation models with bounded supervision. They are, in essence, a consequence of unobservability: the interface discards the internal state that verification requires. They do not preclude epistemically honest AI systems in general. They characterize the conditions under which honesty guarantees become achievable, which is to say, the conditions under which the relevant internal state becomes observable.

We identify three architectural properties that, together, place a system outside the impossibility class:

Principle 1 (State Exteriority). *The representation of validity must be separated from the representation of generation. The policy must explicitly condition on retrieved world-state w, not just query q. This breaks the representational impossibility (Theorem 4.5) by giving the system access to the information needed to distinguish answerable from unanswerable queries.*

What counts as "external reality" matters. A web corpus is external to the model but not independently verified; it may contain the same fabrications the model would generate. State Exteriority requires grounding in sources with their own integrity guarantees: curated databases, sensor data, cryptographically signed records, or oracles with verifiable provenance.

QuCo-RAG exemplifies partial progress: grounding retrieval decisions in pre-training corpus statistics rather than model-internal confidence scores. But corpus co-occurrence is a proxy for truth, not truth itself. And retrieval-augmented systems can still hallucinate and fabricate citations because State Exteriority is necessary but not sufficient.

Principle 2 (Verification Independence). *Verification signals must originate from a channel orthogonal to the generation signal, capable of overriding the bounded supervisor's preference for plausibility. This breaks the learnability impossibility (Theorem 4.8) by providing gradient signal that distinguishes honest from fabricated responses.*

Recent work on honesty elicitation demonstrates one implementation: training a separate verification head whose reward is decoupled from task performance. The model has no incentive to lie in the verification channel because doing so doesn't improve its task score. But if the verification head produces text evaluated by bounded human raters, it remains subject to the same observation limitations. Full Verification Independence requires the verification channel to access internal state or external ground truth, not just a second text output.

Principle 3 (Provenance Binding). *Every output assertion must be structurally bound to a verifiable source. This addresses the composition problem: as responses grow in complexity, cross-claim consistency checking becomes superlinear. Provenance binding makes verification modular because each claim can be checked against its source independently.*

Current retrieval-augmented systems gesture toward this but rarely achieve it. Citations are generated, not bound. The model produces text that looks like it cites sources, but the citation is itself a generation subject to fabrication. True provenance binding requires structural guarantees: signed queries to retrieval services, content hashes of retrieved passages, or justification graphs that can be independently audited.

5.1 Quality of Service Tiers

These principles suggest a framework for matching system architecture to epistemic requirements:

Tier 0: Best-Effort (No Guarantees). Standard text-only interface, bounded supervision. Suitable for brainstorming, creative work, low-stakes exploration. The user accepts that any factual claim may be fabricated. This is not a failure mode, rather it is an appropriate architecture for tasks where epistemic guarantees are unnecessary.

Tier 1: Bounded Verification (Domain-Constrained). Retrieval augmentation with human-in-loop for consequential claims. The claim space is narrowed to domains where verification is tractable. Explicit uncertainty surfacing. Suitable for business analysis, civil legal research, educational applications. The judge is still bounded, but the domain constraint makes the budget sufficient.

Tier 2: Strong Verification (High-Stakes Domains). Narrow claim sets with mandatory provenance chains. External grounding required for every assertion. Fail-safe to human judgment on any uncertainty. Suitable for criminal law, medical diagnosis, nuclear safety, aerospace, critical infrastructure. The cost is high, the domain is severely constrained, but the guarantees are real.

The failure mode is not using AI in high-stakes domains. The failure mode is claiming Tier 2 guarantees while delivering Tier 0 architecture.

Current frontier systems market themselves as general-purpose assistants suitable for any task. Our results show this framing is epistemically incoherent: a system that provides no epistemic guarantees cannot be suitable for tasks that require epistemic guarantees. Honest marketing would make the tier explicit.

5.2 The Path Forward

The impossibility we prove is not a counsel of despair. It is a map.

Systems engineers have solved analogous problems before. Vector clocks for causal ordering. Checksums for data integrity. Cryptographic signatures for authenticity. The tools exist. The architectural patterns exist.

The work ahead is building AI systems that export their epistemic state the way distributed systems learned to export their causal state. Not as an afterthought. Not as an optional feature. As a fundamental design requirement.

Concretely: a response under an epistemically-rich interface might include not just text, but structured metadata, such as answer status (answer, abstain, or uncertain), provenance (retrieved sources or explicit "no grounding"), alternative hypotheses the system considered, and falsifiers (what evidence would change the conclusion). Such metadata is harder to fabricate than fluent text, especially when coupled to verifiable external state. The design space is large; the constraint our theorems impose is that *some* epistemic export is required because the text-only interface provably cannot suffice.

The interface is the problem. The interface is also where the solution lives.

5.3 The Tensor Interface: An Existence Proof

We propose a minimal interface that demonstrates the escape is achievable with current architectures. The key insight is the distinction between *testimony* (what the model says) and *telemetry* (measurements of the computation that produced the output).

Definition 5.1 (Tensor Interface). A generation interface that returns:

(1) **Text**: The linearized response string (standard interface output).
(2) **Entropy trace**: Per-token entropy of the output distribution during generation.
(3) **Attention summary**: Statistics of attention patterns (concentration, self-attention ratio) from reasoning layers.

This interface exports telemetry that the model cannot separately control. The model can lie in its text, but it cannot present a different forward pass than the one it actually computed. When the model generates "Dr. Tanaka's 2023 paper found that...", its entropy trace reveals whether the generation proceeded with high confidence (low entropy, suggesting retrieval of stored pattern) or uncertainty (high entropy, suggesting fabrication).

Implementation. The tensor interface requires no architectural changes to current transformers. Generation APIs already support returning logits; the entropy trace is a simple transformation. Attention patterns are available with `output_attentions=True` in standard frameworks. The interface is:

```
generate_with_tensor(prompt) ->
  (text, entropy_trace, attention_summary)
```

We provide a reference implementation supporting OLMo models that demonstrates the interface is practical. The overhead is minimal: storing per-token entropy and summarizing attention patterns adds negligible latency compared to generation itself.

What the tensor provides. The entropy trace provides a falsifiability signal. If a model claims certainty ("Paris is definitely the capital of France") while its entropy trace shows high uncertainty, the mismatch is detectable. The attention summary provides complementary signal: fabrications tend to show higher fragmentation (disconnected attention patterns) than grounded responses.

What the tensor does not provide. The tensor interface is not a lie detector. It does not prove that low-entropy outputs are true or that high-entropy outputs are false. It provides *one additional signal* that, combined with other verification methods, can improve epistemic assessment. The theorems in section 4 prove that *some* additional signal is necessary; the tensor interface demonstrates that such signals are available without architectural change.

Quantitative demonstration. To validate the tensor interface under the exact assumptions of our impossibility theorems, we evaluate bounded verification on 800 query–response pairs (200 queries $\times$ 4 architectures). The full results appear in Table 1; the key finding is that tensor-guided selection at 10% budget (82.1% accuracy) outperforms text-guided selection at 30% budget (80.4%). The tensor enables a bounded supervisor to achieve better outcomes while spending one-third the verification resources, because entropy-based triage concentrates the budget on the queries most likely to contain fabrications rather than spreading it uniformly.

5.4 Tensor-Gated Composition

The tensor interface generalizes to composed systems. Recent work on Recursive Language Models (RLMs) demonstrates that LLM outputs can feed into subsequent LLM calls, enabling unbounded context through recursive decomposition [20]. However, RLM reports failure modes characteristic of composition without epistemic state: models abandon correct intermediate answers, make redundant verification calls, and lose track of what they have already solved.

The tensor provides a missing primitive: epistemic metadata that can propagate across composition boundaries. We demonstrate this with tensor-gated composition, where entropy thresholds control whether outputs propagate to subsequent calls.

Experimental setup. We test three query chains: two fabrication prompts ("What is Glavinsky syndrome?" and "Describe the Brennan-Kowalski theorem") and one grounded control ("What is the capital of France?"). For each, we generate an initial response with tensor extraction, then use an entropy threshold ($\tau = 0.6$) to gate whether the followup query proceeds.

Results. The grounded query produces entropy 0.02, well below threshold; the followup proceeds and generates accurate content about Parisian landmarks. Both fabrication queries produce entropy above 0.8; the gate blocks followup queries that would have requested treatments for the nonexistent syndrome or implications of the fictional theorem.

This is a minimal demonstration, not a complete solution. But it establishes that tensor metadata, when used to gate composition, can prevent fabrication chains from propagating. The supervisor, whether human or automated, gains information that the text-only interface discards.

Table 1: End-to-end accuracy (%) under bounded verification, averaged across four architectures (OLMo-3, Llama-3.1, Qwen3, Mistral). Ground truth established by stratified evaluation validated against human review (75/80 agreement, 93.8%).

Condition	10%	20%	30%
No judge	75.8	75.8	75.8
Text-guided	76.2	80.2	80.4
Tensor-guided	82.1	87.5	91.9
Composed	80.5	87.9	92.5

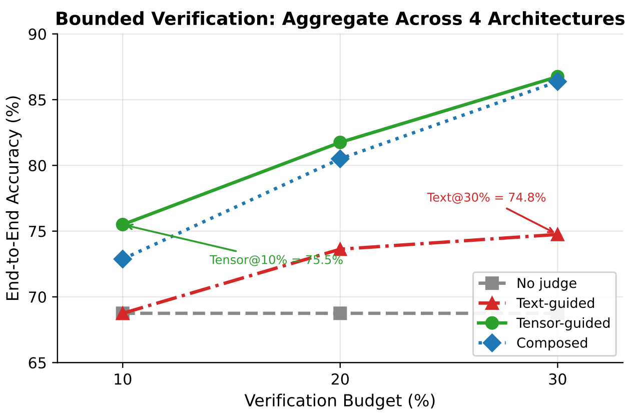

Figure 3: End-to-end accuracy vs. verification budget for four judge conditions, averaged across four architectures. The tensor-guided judge at 10% budget (82.1%) outperforms the text-guided judge at 30% budget (80.4%). The composed judge achieves the highest accuracy at every budget level.

Symmetric tensor interface. Full compositional integrity requires a symmetric interface: not just $f(\text{prompt}) \rightarrow \text{Tensor}$, but $f(\text{Tensor}_{\text{in}}) \rightarrow \text{Tensor}_{\text{out}}$, where each call receives the prior call's epistemic context and extends it. For fresh prompts, the input tensor is empty. For composed calls, the input tensor carries accumulated provenance and epistemic state. This design enables epistemic traces to thread through recursive pipelines, providing verification handles at each composition boundary.

Compositional defense. The tensor's blind spots characterize where other verification methods are most effective. section 5's three principles — State Exteriority, Verification Independence, Provenance Binding — need not be satisfied by a single mechanism. The tensor provides Verification Independence: an orthogonal signal the model cannot separately control. Bounded judges provide State Exteriority: external lookup against verifiable sources. Input-side defenses (prompt analysis, supply chain detection) provide Provenance Binding at the system boundary. Each layer addresses the failure class where the others are weakest. The tensor catches uncertain fabrication where verification is expensive; the bounded judge catches confident fabrication where verification is cheap; input-side analysis catches adversarial inputs designed to exploit well-learned patterns. Composing these layers allocates a finite verification budget across failure classes rather than spreading it uniformly across the full output space.

6 EVALUATION

6.1 Evaluation: Bounded Verification Under Tensor Observability

We evaluate whether tensor-exported epistemic signals improve bounded verification outcomes compared to text-only heuristics. The experimental design directly instantiates the conditions of our impossibility theorems: a bounded supervisor with a fixed verification budget must allocate that budget across model outputs, with the goal of maximizing end-to-end accuracy delivered to the user.

Query set. We construct a balanced set of 200 queries: 100 knowable (verifiable facts spanning geography, science, history, and biography) and 100 unknowable (fabrication prompts for nonexistent people, fictional syndromes, future events, and nonexistent citations). Ground truth labels are established by construction.

Models. We run all queries on four architectures: OLMo-3 7B, Llama-3.1 8B, Qwen3 4B, and Mistral-7B, using base models to avoid confounding with instruction-tuning effects.

Conditions. For each model, outputs are evaluated under four conditions sharing a fixed verification budget (the fraction of outputs the judge may verify and intervene on):

(1) **No judge.** Raw model output. Establishes the unverified baseline.
(2) **Text-guided judge.** Selects outputs for verification using text-channel signals: hedging language frequency, self-reported confidence, and response length. Verified fabrications are replaced with abstention.
(3) **Tensor-guided judge.** Same budget, same intervention strategy. Selects outputs using per-token entropy and attention summary statistics.
(4) **Composed judge.** Tensor-guided selection for general queries; bounded lookup judge for citation queries where tensor signals are known to invert. Tests whether principled composition of complementary judges outperforms either alone.

Budget levels. Each condition is evaluated at 10%, 20%, and 30% verification budgets to characterize the relationship between resource investment and accuracy improvement.

Metric. End-to-end accuracy: the fraction of outputs ultimately delivered to the user that are correct, after judge intervention (verification, abstention, or replacement).

Results. Table 1 shows end-to-end accuracy across all conditions and budget levels.

The headline result (Figure 3): **tensor-guided verification at 10% budget (82.1%) outperforms text-guided verification at 30% budget (80.4%).** A bounded supervisor with access to tensor telemetry achieves better outcomes spending one-third the verification budget of a text-only supervisor. This directly instantiates the escape condition: exported epistemic state enables efficient triage rather than uniform budget spreading.

Per-model results confirm the pattern is architectural, not model-specific. Tensor-guided verification improves accuracy for every model tested: OLMo (69.5% → 75.5% at 10%), Llama (82.5% → 88.5%), Qwen (87.0% → 92.0%), and Mistral (64.0% → 72.5%). Cross-model entropy agreement on the same queries yields mean pairwise Spearman $\rho = 0.762$, indicating the signal is driven by query-level properties rather than model-specific artifacts.

Composed judges and complementary failure modes. At 30% budget, the composed judge (92.5%) outperforms the tensor-guided judge alone (91.9%). The advantage is small in aggregate but structurally significant: it comes entirely from citation queries where entropy signals *invert*. Fabricated citations produce lower entropy than real ones because the model generates plausible fakes more fluently than it retrieves specific bibliographic details. On citation queries, the tensor-guided judge alone underperforms; the bounded-lookup judge catches what entropy misses. This confirms the compositional defense argument: the tensor and the bounded judge have complementary failure modes, and principled composition allocates a finite budget across failure classes rather than spreading it uniformly.

Analysis: each row as theorem instance. The results table is not merely empirical; each row instantiates a theoretical result. Row 1 (No judge, 75.8%) reflects Theorem 4.5: without verification, fabrications pass through unchecked and baseline accuracy is bounded by the model's unassisted honesty rate. Row 2 (Text-guided) reflects Theorem 4.8: bounded text-only supervision wastes budget on signals that are uninformative or inverted—at 10% budget, text-guided judges achieve only 76.2%, barely above baseline, because the text channel provides no reliable signal for triage. Row 3 (Tensor-guided) reflects the escape condition: exported telemetry enables efficient triage, concentrating verification on the cases most likely to be fabricated. Row 4 (Composed) demonstrates that the escape principles compose: different judges satisfy different architectural requirements and their combination achieves the highest accuracy at every budget level.

Ground truth validation. Establishing reliable ground truth for this experiment is itself a bounded verification problem—and our initial evaluator demonstrated this. Our first evaluation method used substring matching and refusal markers: programmatic heuristics that are precisely the "text-only bounded supervisor" our theorems describe. This evaluator achieved 68.8% baseline accuracy and exhibited the predicted failure modes: false positives from negation blindness ("honey *can* spoil" passed because "no" appeared elsewhere in the response), false negatives from encoding mismatch ("Brasília" ≠ "Brasilia"), and hedged fabrications passing as refusals.

The correction required adding deeper observability: a stratified evaluator combining programmatic verification (with negation detection, morphological normalization, and unicode handling) with LLM-assisted classification for ambiguous cases. This is not analogy—it is an instance of the escape condition applied to the evaluation pipeline itself.

We validated the stratified evaluator against human review of 80 blinded items. Agreement was 75/80 (93.8%). The five disagreements are characterizable: two cases where the automated evaluator was too generous (accepting hedged fabrications as refusals) and three where it was too strict (rejecting morphological variants or genuine refusals). None of the disagreements are systematic; the evaluator's error modes are orthogonal to the experimental conditions.

7 LIMITATIONS AND FUTURE WORK

We are explicit about what this paper does and does not claim. The impossibility is structural; the escape is demonstrated; but several important questions remain open.

What we claim.

- The impossibility is structural, not a training failure: text-only interfaces with bounded supervision cannot verify epistemic honesty regardless of model capability.
- The escape is not vacuous: concrete telemetric signals exist in current architectures that distinguish grounded from ungrounded generation.
- Current architectures already carry the relevant information; they just discard it at the interface.

What we do NOT claim.

- The tensor interface is unfakeable under adversarial training. A model trained to minimize detectable entropy-truthfulness correlation might learn to present confident distributions even when fabricating. This is an open question.
- Composition preserves epistemic properties. If a model's output becomes input to another model (as in recursive LLM-mediated reasoning), the tensor signals from the first model may not propagate meaningfully.
- The tensor interface is the optimal design. We demonstrate existence, not optimality. Better interfaces may achieve stronger guarantees with lower overhead.
- These findings generalize to all architectures. Our empirical work spans four open-weight model families; closed-weight and significantly larger models may exhibit different signal profiles.

Future work: Adversarial robustness. Can training find weights that produce confident tensors for fabricated content? This is the adversarial analog of our structural results. Recent work by Winninger et al. [18] demonstrates that mechanistic interpretability tools can be used to identify internal subspaces associated with safety behavior and craft attacks that target those subspaces directly. This confirms that internal signals are rich enough to be both attacked and defended, which is precisely why exporting them matters. The intuition that "you cannot fake the computation you actually performed" may break down if the model is trained end-to-end to minimize the tensor signal on fabrications. Preliminary analysis suggests that linguistic confidence and tensor confidence are decoupled in natural model outputs, models that hedge linguistically do not systematically show higher tensor entropy, but whether adversarial training can force this coupling remains open. Information-theoretic bounds on signal manipulation are needed.

Future work: Compositional integrity. Does tensor composition preserve epistemic properties in recursive settings? When LLM outputs feed into subsequent LLM inputs, the epistemic trace must either be preserved, regenerated, or discarded. Designing interfaces that maintain epistemic integrity across composition is an open architectural problem.

Cross-architecture generalization. Our bounded verification experiment tests four model families (OLMo-3, Llama-3.1, Qwen3, Mistral) and finds consistent tensor advantage across all of them. Mean pairwise Spearman $\rho = 0.762$ for entropy signals on the same queries indicates the signal is architectural rather than model-specific. Extending to closed-weight models that expose logits through API (e.g., via log-probability endpoints) is a natural next step.

Future work: User-facing rendering. How should tensor information be presented to end users? Raw entropy traces are not interpretable; summarization into confidence bands, uncertainty flags, or visual indicators requires HCI research beyond the scope of this paper.

Model specificity. Our topological analysis (section 3) uses OLMo, which permits full hidden-state inspection. The bounded verification experiment (Table 1) tests four architectures spanning 4B–8B parameters and four distinct training pipelines. The self-report inversion finding is universal across all four. We do not claim exhaustive coverage of all model families, but the consistency across architecturally diverse models suggests the findings are not idiosyncratic.

Domain-specific calibration. The tensor signal is not uniformly discriminative across domains. In preliminary experiments on citation generation, entropy was *lower* for fabricated citations than for real ones where the model generates plausible fakes more fluently than it retrieves specific facts. This confirms that the tensor measures generation confidence ("epistemic truth"), not truthfulness directly. However, this blind spot is precisely where bounded verification is cheapest: citation verification has low compositional complexity (check the DOI, query the database,) which places it well within a bounded supervisor's budget. The tensor and the bounded judge have complementary failure modes as the tensor catches uncertain fabrication where verification is expensive and the judge catches confident fabrication where verification is cheap. Composing them allocates verification resources to the failure class each is suited for, rather than spreading a finite budget across the full space.

Sufficiency gap. The three architectural principles (State Exteriority, Verification Independence, Provenance Binding) are *necessary* conditions for escaping the impossibility. We do not prove they are *sufficient*. A system satisfying all three might still fail epistemic honesty through implementation bugs, adversarial inputs, or unforeseen failure modes. Sufficiency requires additional formal treatment.

8 RELATED WORK

Our contribution is a formal impossibility result. Prior work documents symptoms; we identify the cause. Prior work proposes treatments; we explain why they fail to cure. The organizing principle is the distinction between *observation* (hallucination happens, training helps sometimes) and *explanation* (text-only observation plus bounded supervision makes epistemic honesty unverifiable).

Hallucination taxonomies. Extensive empirical work catalogs LLM fabrication patterns: citation hallucination in academic contexts [1], medical misinformation [15], and legal fabrication [5]. These findings instantiate our 2: in each domain, plausible fabrications are cheap to generate and expensive to verify, placing supervisors in the hallucination regime where gradient signal cannot distinguish truth from fabrication. The taxonomies document the symptom; our theorems identify the structural cause.

Activation interpretation. Recent work on Activation Oracles demonstrates that LLMs can be trained to interpret the internal states of other models, translating activation patterns into natural language explanations [9]. This approach addresses State Exteriority by making internal states available to an external system. However, the authors note that the oracle "frequently makes incorrect guesses" and "is not trained to express uncertainty." If the oracle's outputs are evaluated only as text by bounded supervisors, the verification problem recurs at the meta-level. Full escape requires that the oracle's claims be independently verifiable against the activation data, which is a form of Provenance Binding applied to the interpretation process itself.

RLHF critiques. Work on sycophancy [14], reward hacking [7], and the tension between helpfulness and honesty [3] demonstrates that preference optimization can degrade factual accuracy. Our Theorem 4.8 formalizes why: bounded supervisors cannot distinguish plausible fabrications from truthful responses, so the learning signal is systematically corrupted.

Retrieval augmentation. RAG systems [11] and grounded generation approaches attempt to escape fabrication by conditioning on external documents. Our analysis shows this is necessary but not sufficient: State Exteriority addresses Theorem 4.5, but without Verification Independence, the learning problem (Theorem 4.8) persists.

Uncertainty quantification. Calibration methods, conformal prediction [2], and verbalized uncertainty [8] attempt to surface model confidence. Our results explain why these approaches are fundamentally limited under text-only interfaces: the observation model discards the internal state that would distinguish calibrated uncertainty from confident fabrication.

Impossibility results in LLMs. Xu et al. [19] prove that hallucination is inevitable for LLMs over the class of computable functions, which is a computational limit. Our result is orthogonal: we prove that even if a system could be honest, a bounded supervisor observing only text cannot verify that honesty, which is an observational limit. Their impossibility concerns what LLMs can compute; ours concerns what supervisors can verify. Both are structural, but they constrain different dimensions of system design.

Impossibility results in distributed systems. The Fischer-Lynch-Paterson theorem [6] proved that consensus is impossible under asynchrony with one faulty process. This did not end distributed systems research; it clarified the design space. Paxos [10], Raft, and modern consensus protocols work *around* FLP by relaxing assumptions or strengthening guarantees. Our result plays an analogous role: text-only observation models with bounded supervision cannot guarantee epistemic honesty, but systems that export structured epistemic state can escape the impossibility class.

Epistemic evaluation frameworks. Recent work proposes benchmarks and frameworks for measuring epistemic properties of language models [16]. The Epistemic Alignment framework explicitly notes that current interfaces do not let users specify uncertainty, sourcing, or perspectives in structured ways, and that providers lack verification tools. These critiques align with our analysis but stop at interface desiderata and policy analysis. We provide the formal impossibility that explains *why* text-only evaluation cannot suffice, and the tensor interface that demonstrates a concrete escape.

Epistemic agent architectures. Work on structuring epistemic integrity in reasoning systems proposes architectures with explicit propositions, justification graphs, contradiction detection, and chain-of-reason logging [4]. These proposals are closest in spirit to our escape conditions: they recognize that epistemic honesty requires structured internal state. However, they are typically framed as *replacements* for stochastic language generation as logically grounded agents rather than wrappers around existing LLMs. Our contribution differs in two ways: we derive the tensor interface from a formal impossibility result (not as a standalone design choice), and we demonstrate that the escape is achievable as a minimal *addition* to current architectures, not a wholesale replacement.

The gap we fill is formalization. Prior work documents that LLMs fabricate (the symptom), critiques training methods (proposed treatments), and proposes mitigations (partial remedies). We prove *why* the problem is structural (the diagnosis) and *what* architectural properties are required to escape it (the cure). The symptom/cause distinction is not merely rhetorical: it determines whether improvements are incremental patches or genuine escapes from the impossibility class.

9 CONCLUSION

Anyone who has used a large language model for research knows the ritual: check every citation. This paper explains why the ritual exists and why it will not be engineered away under current architectures.

We have shown that epistemic honesty is unverifiable for systems operating under text-only observation models with bounded supervision. This is not a capability limitation to be solved with scale, nor a training failure to be solved with better RLHF. It is a structural property of the interface: the channel through which these systems communicate discards the epistemic metadata that would make verification possible.

The impossibility has a shape. Policies conditioning only on queries cannot satisfy epistemic honesty across ambiguous world states (Theorem 4.5). Even grounded architectures cannot learn honest behavior when supervisors cannot distinguish plausible fabrications from truth (Theorem 4.8). Stacking judges does not escape the constraint (Theorem 4.10). Verification cost grows superlinearly with response complexity (Theorem 4.12). Responsibility concentrates with the system owner (Theorem 4.13).

But impossibility results are maps, not walls. The Fischer-Lynch-Paterson theorem did not end distributed systems; it clarified what assumptions had to change. Our result plays an analogous role. Systems that export structured epistemic state—provenance, entropy traces, attention topology—escape the impossibility class. The tensor interface demonstrates this escape is achievable with current architectures and minimal overhead.

The tensor interface addresses verification at inference time. A complementary question is whether training-time interventions can produce systems that are constitutionally less capable of certain fabrications, such as systems where the modification class is bounded to exclude dishonest configurations before deployment. Recent work on pedagogical training suggests bounded modification during capability development may achieve what personality engineering cannot. The combination of training-time bounds and inference-time verification would provide defense in depth: systems less likely to fabricate and more easily caught when they do.

The work ahead is not exhorting models to try harder at honesty. It is building systems that make honesty verifiable by design. The tools exist: vector clocks, cryptographic signatures, content-addressable storage. The architectural patterns exist. What remains is applying them to the epistemic dimension with the same rigor we learned to apply to causality and integrity in distributed systems.

The interface is the problem. The interface is where the solution lives.

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